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The impact of variation and autonomy on psychological responses to high intensity interval training exercise☆

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ABSTRACT

Introduction: High-intensity interval training (HIIT) provides notable physiological benefits and is generally well-tolerated across modalities and populations. This study investigated how exercise autonomy support impacts psychological responses to exercise.
Methods: Twenty-nine participants completed three HIIT trials: Conventional-HIIT with 60-sec work segments, Varied-HIIT with a mix of 30, 60, 90, & 120-sec segments, and Autonomous-HIIT with self-selected 30, 60, 90, & 120-sec segments. Affective valence, enjoyment, and intention were measured.
Results: Affective valence during exercise was not different between trials ($p > 0.05$) but enjoyment during exercise was higher for Autonomous-HIIT ($p < 0.05$). Enjoyment and intention measured post-exercise were greater for Autonomous-HIIT than Varied-HIIT ($p < 0.05$).
Conclusion: Autonomous HIIT produced more desirable responses than varied and traditional HIIT sessions. These data suggest that HIIT sessions utilizing self-selected interval durations can produce more positive responses, which provides the basis for recommending autonomy within HIIT exercise.

1. Introduction

Physical activity is an established intervention for cardiometabolic disease treatment and prevention, and is also known to improve mental health (Das & Horton, 2012). Despite these many benefits, epidemiological data indicates that one in four adults worldwide do not meet the current physical activity guidelines (Foster et al., 2018). These guidelines recommend at least 150 min of moderate-intensity physical activity or 75 min of vigorous-intensity physical activity each week (ACSM, 2018). Given that the relationship between amount of physical activity and health-related outcomes follows a dose response pattern (Piercy & Troiano, 2018), any increase in physical activity above baseline can produce changes that positively impact health and wellness. The increased understanding that numerous exercise modalities, intensities, and durations can provide health benefits has encouraged leading organizations to begin promoting approaches to physical activity that are more innovative. One notable example is that the American College of Sports Medicine now formally acknowledges interval-based exercise as an approach to exercise that should be considered a viable strategy for meeting physical activity recommendations (ACSM, 2018).

Utilization of high-intensity interval training (HIIT) in particular has become increasingly popular in recent years (Batacan et al., 2017) and continues to be included on lists of worldwide fitness trends (Thompson, 2019). HIIT exercise is highly variable in terms of intensity, duration, and configuration, but collectively this type of exercise session is now recognized as a reasonable approach for improving health and fitness. Specific physiological benefits of HIIT have been observed for a variety of cardiovascular and metabolic indicators and across both clinical and general populations (Batacan et al., 2017; Gibala et al., 2012; Viana et al., 2019). Concerns regarding the tolerability of HIIT have been raised (Coyle, 2005), but research makes clear that a wide variety of

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HIIT protocols produce psychological responses that are perhaps equal to and in some cases more positive than the continuous moderate-intensity exercise that is most often prescribed for health and wellness (Oliveira et al., 2018; Stork et al., 2017).

Primary variables of interest within HIIT studies are affective valence and enjoyment. Though these constructs are similar, they differ in that affective valence arises without significant thought or cognitive elaboration (Ekkekakis & Russell, 2013), while enjoyment includes more cognition and interpretation of the exercise experience (Wankel, 1993). Affective valence in particular has received increased attention because research evidence indicates that in-task affective valence is more predictive of future exercise behavior and adherence than measures of post-exercise affective valence (Rhodes & Kates, 2015), though recent findings from a pilot study that included HIIT did not demonstrate in-task affect as predictive of future exercise behavior (Ivanova, et al., 2020). This line of inquiry is valuable because it has been estimated that a one-unit positive increase in affective valence is associated with 15–30 min of additional exercise per week (Williams et al., 2012). This area of literature makes clear that affective responses are highly dependent upon exercise intensity (Ekkekakis et al., 2000). Specifically, the Dual-Mode Theory suggests that affective valence is generally steady and positive as exercise intensity increases up to the anerobic threshold and begins to decline and/or become negative beyond the anaerobic threshold through peak aerobic exercise work capacity (Ekkekakis et al., 2005). Importantly, research investigating the psychological responses to HIIT suggest that affective and enjoyment responses are generally positive (Oliveira et al., 2018; Stork et al., 2017), possibly because the high-intensity component of HIIT may be sufficiently brief so as to not provide adequate time for perceptions of the experience to become negative (Stork et al., 2017). Though specific characteristics of interval intensity, duration, and recovery are important considerations in the prescription of HIIT, the body of evidence is generally supportive of HIIT as a health and fitness strategy.

One additional consideration regarding the exercise experience that has received increased interest within the exercise psychology literature within the context of acute exercise is autonomy. This construct is a major component of Self-Determination Theory (SDT), which describes autonomy as a basic and innate need that impacts human behavior (Ryan & Deci, 2000). SDT suggests that motivation towards a behavior is enhanced when individuals perceive personal agency and autonomy regarding their actions, while perceptions that behavior is being manipulated or controlled by others leads to reductions in autonomy. Further, SDT predicts that behaviors undertaken autonomously result in greater enjoyment, more positive affective responses, and adherence (Rose & Parfit 2012). An aspect of autonomy that facilitates a desirable motivational state within the context of exercise is revealed when participants are able to self-select and participate in the development of exercise sessions. In contrast, autonomy and motivation would be expected to be jeopardized when exercise is prescribed or imposed, especially if done so in a manner that is perceived as authoritative or controlling (Deci & Ryan, 1987). Research in this area is limited but does indicate that self-selected exercise leads to more positive affective valence during exercise (Oliveira et al., 2015). Autonomy during exercise is still an emerging area of the literature and there are several ways to implement it within exercise sessions and training programs. Broadly it could be as simple as choosing what mode of exercise to participate in for a specific session, while a more narrow approach to autonomy could include choosing to make changes to intensity during an exercise session. Several investigations aimied to directly consider how reductions in autonomy impact motivation demonstrated that a self-selected session of continuous exercise produced greater perceived choice, enjoyment, interest, and affective valence when compared to an imposed session (Ekkekakis et al., 2010; Parfitt et al., 2012). Further elaboration of this point is provided by research demonstrating imposed exercise intensities greater than that which is self-selected results in less positive affective valence responses (Lind et al., 2008; Williams et al., 2015). An insightful study comparing self-selected and imposed moderate continuous exercise completed at the same intensity revealed that imposed exercise produced less positive affective valence than self-selected exercise despite both sessions being equivalent in terms of workload (Hamlyn-Williams et al., 2014).

Additional support for the importance of autonomy on motivation and behavior beyond acute exercise experiences is available across a variety of contexts. Research indicates fitness facilities that provide a climate that supports autonomy creates greater intrinsic motivation, commitment, and enjoyment (Fortier et al., 2012; Moore & Fry, 2017). Similarly, in rehabilitation contexts the provision of autonomy in the selection of therapeutic exercises increase self-determined motivation and rehabilitation adherence when compared to a more controlled motivation (Chan et al., 2009; Russell & Bray, 2010). Finally, it has been demonstrated within physical education environments that students who participate in the selection of their physical fitness sessions exhibit higher levels of intrinsic motivation and physical activity intention (Sfandyari et al., 2020).

The literature to date that has considered the impact of self-selection on motivational and affective responses is valuable and supportive of the importance of autonomy and self-selection, but much more work is needed including consideration of the impact of fitness or physical activity experience. Importantly, only one study to date has investigated these issues within the context of HIIT (Kellogg et al., 2019). This research compared HIIT that was imposed at a fixed percentage of peak workload against a HIIT session where workload was self-selected and corresponded to a prescribed RPE. Findings from this study indicate that self-selected HIIT produced higher oxygen consumption, perceived exertion, and lower affective valence. Importantly, interpreting the impact of autonomy is limited because the two exercise sessions were not equal on work and because the autonomous condition included instructions for participants to exercise at a specific level of exertion, thus limiting the amount or type of autonomy experienced by the exerciser. Research to date within HIIT exercise has also not yet considered the possibility that variation of interval length throughout a session may be adequate to produce more desirable affective or motivational responses. Variation has been shown to stimulate interest and enjoyment and in the case of exercise has also been shown to predict future behavior (Sylvester et al., 2016). Outside of HIIT, perceived exercise variety has been indirectly and positively associated with exercise behavior via autonomous exercise motivation (Sylvester et al., 2016). Additional support for this possibility is provided by research demonstrating that provision of an increased expectation for variety within a group exercise class creates greater enjoyment and interest than a class that did not include the expectation of variety even though the exercise sessions were equivalent (Magaraggia et al., 2013).

Given that utilization of interval-based exercise continues to be a common aspect of health and fitness enhancement, further evaluation of the psychological characteristics of HIIT exercise sessions is warranted. Importantly, some exercisers may not be attracted to conventional HIIT sessions that repeat the same interval intensity and duration throughout the exercise session, though it is possible that the simplicity and limited thought required to participate in prescribed HIIT exercise may be preferable for some. The present study aims to investigate the effect of allowing exercisers to experience HIIT in ways that are variable and includes differing levels of autonomy. Based on SDT it is hypothesized that exercisers given more autonomy will have more positive psychological responses compared to imposed HIIT sessions despite each session of exercise being matched for time, intensity, and total work.

2. Methods

2.1. Participants and research design

Participants were 29 adults (16 males, 13 females, mean age ± SD = 28 ± 7 years; mean BMI ± SD = 26 ± 3), from a large university and surrounding community in the southeastern United States. Inclusion criteria were: ages 18–45 years, BMI of 18–35, and participation in 150 min per week of physical activity for the previous three months. An a priori power analysis using G*power (Faul et al., 2007) for repeated-measures, within-factors analysis of variance ANOVA provided guidance on the required sample size likely required to detect significant differences. Anticipating an approximated medium eta squared effect size of 0.06 for affective valence based on published research (Lind et al., 2008), desired statistical power of 80%, a Type 1 error rate of 5%, a violated assumption of sphericity set at an epsilon of 0.7, and a correlation of repeated measures of 0.6, yielded a required sample of 24 participants. A total of 34 participants were recruited to allow for 30% dropout. Five participants (15%) dropped out and the remaining 29 participants are included in all analyses. Given the novelty of the design, utilization of this convenient sample represents a reasonable first study of this particular topic. Participants were required to complete five laboratory visits, each visit separated by at least 24 h, and visits were generally completed over a period of one to four weeks. The first visit included study paperwork and maximal exercise testing. The second visit was meant to provide participants with an opportunity to become familiarized with laboratory equipment, measurement scales, and protocols. The remaining three visits served as the experimental sessions. All procedures were approved by the university institutional review board.

2.2. Screening

Screening included the completion of informed consent, health history questionnaire, and measurement of height, body weight, heart rate (HR), and blood pressure (BP). Participants were instructed to avoid alcohol, caffeine, and tobacco for 3 h prior to testing (ACSM, 2010). Body composition was assessed by way of bioelectrical impedance (InBody, Cerritos, CA).

2.3. Metabolic testing

A progressive, ramp protocol was performed on an electronically braked cycle ergometer (Lode, Groningen, Netherlands). The protocol ramp rate varied between 15 and 20 W per minute for females and 20–25 W per minute for males based on self-reported descriptions of exercise history and experience. The test progressed until volitional fatigue as evidenced by the participant not being able to maintain a pedal cadence of 50 revolutions per minute. A 3-min cool-down immediately followed peak exercise and test termination. Heart rate (HR), blood pressure (BP), ratings of perceived exertion (RPE), and expired gases were monitored in accordance with standard exercise testing guidelines (ACSM, 2010). HR was measured using a HR monitor (Polar, Bethpage, NY), BP was determined by auscultation, and RPE was estimated each minute using the Borg 6–20 scale (Borg, 1998). Expired gases were collected using a neoprene mask and analyzed continuously using a calibrated metabolic measurement cart (Medical Graphics, St. Paul, MN). Peak oxygen consumption (VO2 peak) was identified as the largest volume of oxygen consumed per minute during the test. Criteria for verifying maximal exertion were as follows: a peak HR of at least 90% of age-predicted maximal HR (based on 220 – age), a peak RPE of at least 18 (on a 6–20 scale), and a peak respiratory exchange ratio (RER) of at least 1.15.

2.4. Familiarization

The objective of the familiarization session was to give participants an opportunity to experience the varying work interval durations associated with the experimental sessions, namely 30, 60, 90, and 120 s. This process included allowing participants to experience each workload two times so as to obtain a clear sense of the time and effort associated with each interval duration. Participants were also familiarized to all experimental data collection procedures, scales, and questionnaires used during data collection. After completion of the familiarization session participants were asked to build a self-selected HIIT experimental session totaling 20 min including their preferred number of each interval duration. This step within the design allowed participants the opportunity to create a session of HIIT exercise that reflected their preferences and thus created an autonomous experimental trial.

2.5. Experimental sessions

Each experimental session was conducted identically on an electronically braked cycle ergometer (Lode, Groningen, Netherlands), with the only difference being the specific characteristics of the intervals included within the HIIT session. All sessions were randomized using the balanced latin square technique. Upon arrival participants were seated for 5 min before completing baseline assessments. Participants completed a 3-min warm-up prior to the HIIT session and a 2-min cool-down upon completion of the HIIT session. The experimental exercise sessions were matched for total duration (20 min), intensity of the work interval (90% peak work capacity), intensity of the recovery interval (10% peak work capacity), and total time spent in the work and recovery phases of the session (10 min each). Control of these variables yielded sessions of exercise that were equal on total work completed. The sessions varied only on the configuration of the duration of intervals within the sessions. The result of these design elements was the creation of two prescribed sessions and one self-selected session. Conventional HIIT (C-HIIT) consisted of prescribed 60-s segments of work and recovery intervals repeated 10 times. Varied HIIT (V-HIIT) consisted of prescribed varied-duration intervals done in the order of; 60-, 90-, 30-, 120-, 90-, 60-, 120-, and 30-s intervals. Autonomous HIIT (A-HIIT) consisted of self-selected intervals selected during the familiarization session that included on average approximately five 30-s intervals, four 60-s intervals, one 90-s interval, and one 120-s interval.

2.6. Instrumentation

Variables of interest in the current study were: exertion, affect, enjoyment, and intention for future exercise. Exertion was assessed using the single-item, 15-point (6: "No Exertion At All" to 20: "Maximal Exertion") Borg Scale (Borg, 1998). Affective valence was assessed using the single-item, 11-point (−5: "Very Bad" to +5: "Very Good") Feeling Scale (Hardy & Rejeski, 1989). In-task enjoyment was assessed using the single-item, seven-point (1: "Not At All" to 7: "Extremely") Exercise Enjoyment Scale (Stanley & Cumming, 2010). Post-exercise enjoyment was assessed using the Physical Activity Enjoyment Scale (Kendzierski & DeCarlo, 1991). The PACES is an 18-item, seven-point bipolar rating scale that utilizes the stem "Please rate how you feel at this moment about the exercise you have been doing" and ranges from 1 to 7, with a minimum total score of 18 and a maximum total score of 126. This scale demonstrated excellent internal consistency in the current study (Cronbach's alpha = 0.88 to 0.90). Anchors are provided by two contrasting statements and respondents were asked to indicate strength of agreement. Intention for future exercise was assessed using a single-item, seven-point scale (1: "Very Unlikey" to 7: "Very Likely") that was developed for use in the current study.

2.7. Data collection

Affect was assessed before and during exercise, while enjoyment was assessed during and after exercise. Baseline affect was assessed immediately after the participant was provided with a description of the upcoming session, approximately 5 min before the start of exercise. In-task RPE, affect and enjoyment were assessed eight times during the interval sessions (four times each for the work and recovery phases). All assessments occurred during the last 15 s of the work and recovery intervals most proximal to 25%, 50%, 75%, and 100% of session completion. Post-exercise enjoyment and intention were assessed 5 min after completion of exercise. All assessments taken outside of the exercise session occurred while seated comfortably in a reclining chair in the same room as the exercise equipment. All assessments during the exercise session were taken by asking the participant to verbalize their perceptions while being provided with the scale as visual reference. HR was recorded at the same time points as RPE, affect and enjoyment. Workload changes were controlled by software linked to the testing system. Interactions between the research staff and participant were mostly limited to required data collection and members of the research staff remained mostly out of view of the participant during sessions. To ensure each session was conducted under the same condition, all sessions were completed in a lab setting to eliminate environmental factors such as climate and scenery, and each participant completed all their sessions around the same time of day.

2.8. Statistical analysis

Data was analyzed using SPSS 26 and proceeded in two phases. The first phase included a descriptive analysis of the sample and characteristics of the exercise sessions. Remaining phases utilized repeated measures, within subjects design analyses. The primary analysis included 3 (Session: C-HIIT, A-HIIT, V-HIIT) x 4 (Time: 25, 50, 75, 100%) repeated measures ANOVAs for each dependent variable for both work and recovery segments, namely HR, FS, RPE, and EES. A grand mean in-task value for each dependent variable was also calculated and analyzed using a one-way ANOVA. Post-exercise PACES and intention were also analyzed using a one-way ANOVA. Dependent t-tests were used to determine specific differences observed within the ANOVA. All follow-up testing used exercise session as the within subjects measure. Criterion for significance was set at $p < 0.05$ for all results and exact p-values are reported to facilitate consideration of potential alpha error inflation. Cohen's d is the effect size (ES) reported for mean differences (Cohen, 1992).

3. Results

3.1. Descriptive data

Participants were measured to have a peak VO_2 of 35.1 ± 6.1 mL kg^{-1}•min^{-1}, a maximal HR of 179 ± 10 beats•min^{-1}, a maximal RPE of 19.1 ± 1.0, and a maximal respiratory exchange ratio of 1.26 ± 0.07. Collectively, these results suggest that maximal effort was likely achieved. Specifically, 96% of participants (n = 28) reached the criterion for maximal effort (RPE ≥18), 85% of participants (n = 25) reached the criterion for maximal heart rate (HR ≥ 90% age-predicted maximum), and 96% of participants (n = 28) reached the criterion for maximal respiratory exchange ratio (RER ≥1.15). All participants (n = 29) reached two of the criteria and a strong majority (n = 21) reached all three criteria for maximal effort during metabolic testing.

3.2. Heart rate during exercise

Significant effects were observed for time (F = 18.01, df = 3,0, $p < 0.001$), session (F = 6.58, df = 2,52, p = 0.003), and the interaction of time and session (F = 21.58, df = 6,0, $p < 0.001$) for work intervals. Significant effects were also observed for time (F = 187.85, df = 3,0, $p < 0.001$), session (F = 11.64, df = 2,0, $p < 0.001$) and interaction of time and session (F = 26.98, df = 6,0, p = 0.001) for recovery intervals. These results indicate that HR increased from the beginning to the end of the sessions for work and recovery intervals, and that HR varied between sessions for the work interval. Analysis of the average HR values (C-HIIT = 163 beats/min; V-HIIT = 161 beats/min; A-HIIT = 157 beats/min) for work intervals revealed a significant session effect (F = 6.58, df = 2,52, p = 0.003). Dependent t-tests indicated that the average work interval HR was higher for C-HIIT than A-HIIT (t = 3.60, p = 0.001, ES = 0.65). Evaluation of HR data relative to maximal values indicate that work interval HR was approximately 90% max, which suggests that the interval manipulation was successful in creating a physiologically stressful session of HIIT exercise.

3.3. Perceived exertion during exercise

A significant main effect was observed for time (F = 30.99, df = 3,78, $p < 0.001$), session (F = 3.20, df = 2,52, p = 0.049), and the interaction of time and session (F = 12.07, df = 6,156, $p < 0.001$) for work intervals. Significant effects were also observed for time (F = 8.40, df = 3,81, $p < 0.001$), but not for session (F = 8.40, df = 3,81 p = 0.822) or the interaction of time and session (F = 1.25, df = 6,162, p = 0.292) within recovery intervals. These results indicate that RPE increased from the beginning to the end of the sessions for work and recovery intervals, and that RPE varied between sessions for the work interval. Analysis of the average RPE values for work intervals revealed a significant session effect (F = 3.20, df = 2,52, p = 0.049) and these findings are presented in Table 1. Dependent t-tests indicated that the average work interval RPE was higher for C-HIIT than A-HIIT (t = 2.55, p = 0.017, ES = 0.48).

3.4. Affective valence during exercise

Significant effects were observed for time (F = 6.00, df = 3,78, p = 0.001) and the interaction of time and session (F = 4.27, df = 6,156, p = 0.006) but not for session (F = 2.80, df = 2,52, p = 0.070) for work intervals. Significant effects were also observed for time (F = 9.22, df = 3,81, $p < 0.001$), but not session (F = 1.86, df = 2,54, p = 0.17) or the interaction of time and session (F = 1.33, df = 6,162, p = 0.26) for recovery intervals. These results indicate that in-task affective valence changed over time within sessions. Affective valence did not vary between sessions for the work or recovery intervals, though there was a trend towards significance for the work intervals. Dependent t-tests indicated that the average affective valence was higher for A-HIIT than V-HIIT for work intervals (t = 2.97, p = 0.006, ES = 0.57) and recovery intervals (t = 2.26, p = 0.032, ES = 0.42). See Table 2.

3.5. Enjoyment during exercise

Significant effects were observed for time (F = 4.49, df = 3,78, p = 0.006), session (F = 4.18, df = 2,52, p = 0.021), and the interaction of time and session (F = 3.82, df = 6,156, p = 0.001) for work intervals. Significant effects were also observed for time (F = 10.17, df = 3,81, $p < 0.001$) for recovery intervals but not for session (F = 2.31, df = 2,54, p =

Table 1
Exertion responses as measured by the borg RPE scale.

	Conventional (C-HIIT)	Varied (V-HIIT)	Autonomous (A-HIIT)
Interval Work			
25% Complete	13.9 ± 1.9	14.1 ± 2.2	13.6 ± 1.7
50% Complete	15.2 ± 1.3	16.5 ± 1.9	14.4 ± 1.9
75% Complete	16.2 ± 1.4	15.4 ± 2.0	15.4 ± 1.7
100% Complete	16.3 ± 1.6	14.3 ± 2.1	15.5 ± 2.3
Average	15.3 ± 1.2[a]	15.1 ± 1.7	14.8 ± 1.5
Interval Recovery			
25% Complete	10.6 ± 2.0	10.2 ± 1.9	10.6 ± 2.0
50% Complete	10.8 ± 2.3	10.1 ± 2.0	10.7 ± 2.3
75% Complete	11.4 ± 2.7	11.5 ± 2.5	11.3 ± 2.5
100% Complete	11.0 ± 2.6	11.0 ± 2.4	10.7 ± 2.6
Average	10.9 ± 2.2	10.7 ± 1.9	10.9 ± 2.1

Exertion response for all exercise conditions. Data are presented as mean ± standard deviations.

[a] Exertion was higher for C-HIIT than A-HIIT during interval work (p = 0.017).

Table 2
Affective valence responses as measured by the feeling scale.

	Conventional (C-HIIT)	Varied (V-HIIT)	Autonomous (A-HIIT)
Pre-Exercise	3.3 ± 1.5	3.2 ± 2.0	3.3 ± 1.3
Interval Work			
25% Complete	2.3 ± 1.0	2.0 ± 1.7	2.3 ± 1.2
50% Complete	1.8 ± 1.5	0.9 ± 1.9	2.1 ± 1.5
75% Complete	1.6 ± 1.7	1.2 ± 2.0	1.9 ± 1.6
100% Complete	1.8 ± 2.0	2.3 ± 1.5	2.0 ± 2.3
Average	1.8 ± 1.4	1.6 ± 1.4	2.1 ± 1.4[a]
Interval Recovery			
25% Complete	3.1 ± 0.9	3.1 ± 1.0	3.1 ± 1.0
50% Complete	2.6 ± 1.2	2.7 ± 1.8	2.8 ± 1.2
75% Complete	2.4 ± 1.4	1.9 ± 1.8	2.6 ± 1.4
100% Complete	2.8 ± 1.6	3.0 ± 1.4	3.2 ± 1.4
Average	2.7 ± 1.1	2.6 ± 1.2	2.9 ± 1.1[a]

Affective response for all exercise conditions. Data are presented as mean ± standard deviations.
[a] Affective valence was higher for A-HIIT than V-HIIT during work (p = 0.006) and recovery (0.032).

0.109) or the interaction of time and session (F = 0.51, df = 6,162, p = 0.797) within recovery intervals. Evaluation of these results indicate that enjoyment across work segments varied based on session type, which included interval work segments of different duration. Analysis of the average enjoyment values for work intervals revealed a significant session effect (F = 4.18, df = 2,52, p = 0.021). Dependent t-tests indicated that the average enjoyment was higher for A-HIIT than C-HIIT (t = 2.07, p = 0.048, ES = 0.39) and V-HIIT (t = 3.04, p = 0.005, ES = 0.59). See Table 3.

3.6. *Enjoyment and intention post-exercise*

Evaluation of responses post-exercise included enjoyment and intention. A main effect was not observed for enjoyment (F = 2.71, df = 2,56, p = 0.076) or intention (F = 2.86, df = 2,56, p = 0.066), but observed differences did trend towards significance. Given these trends, dependent t-tests were conducted to identify any differences between individual sessions and indicated significant differences during the post-exercise period. Specifically, enjoyment was greater for A-HIIT than V-HIIT (t = 2.44, p = 0.021, ES = 0.49) and intention was greater for A-HIIT than V-HIIT (t = 2.16, p = 0.030, ES = 0.42). See Table 4.

Table 3
Enjoyment responses as measured by the exercise enjoyment scale.

	Conventional (C-HIIT)	Varied (V-HIIT)	Autonomous (A-HIIT)
Interval Work			
25% Complete	3.9 ± 1.1	3.4 ± 1.0	3.9 ± 1.1
50% Complete	3.7 ± 1.4	3.2 ± 1.5	4.2 ± 1.1
75% Complete	3.5 ± 1.7	3.5 ± 1.4	3.9 ± 1.5
100% Complete	3.8 ± 1.7	4.2 ± 1.4	4.3 ± 1.6
Average	3.7 ± 1.4	3.6 ± 1.2	4.0 ± 1.2[a]
Interval Recovery			
25% Complete	4.3 ± 1.0	4.1 ± 0.8	4.3 ± 1.0
50% Complete	4.2 ± 1.0	4.1 ± 1.0	4.4 ± 1.0
75% Complete	4.2 ± 1.3	3.9 ± 1.2	4.2 ± 1.2
100% Complete	4.7 ± 1.4	4.4 ± 1.4	4.9 ± 1.2
Average	4.3 ± 1.1	4.1 ± 0.9	4.4 ± 1.0

Enjoyment responses for all exercise conditions. Data are presented as mean ± standard deviations.
[a] Enjoyment was higher for A-HIIT during interval work than V-HIIT (p = 0.005) & C-HIIT (p = 0.048).

Table 4
Post-exercise responses.

	Conventional (C-HIIT)	Varied (V-HIIT)	Autonomous (A-HIIT)
Enjoyment	96 ± 15	92 ± 14	*98 ± 13
Intention	3.8 ± 1.8	3.4 ± 1.7	*4.0 ± 1.8

Post exercise responses for all exercise conditions. Enjoyment as measured by the Physical Activity Enjoyment Scale. Data are presented as mean ± standard deviations.
* Enjoyment was higher for A-HIIT than V-HIIT (p = 0.021)
* Intention was higher for A-HIIT than V-HIIT (p = 0.040)

4. Discussion

The purpose of the present experiment was to investigate how autonomy support and variation affect psychological responses to HIIT exercise amongst young, healthy, and active adults. The design included three sessions of HIIT each lasting 20 min in duration alternating between 90% of max work and 10% of max work with a 1:1 work-to-recovery ratio. The variable manipulated during the sessions was the length of work and recovery intervals, utilizing a conventional prescribed 60-s interval duration (C-HIIT), interval durations that were not prescribed but varied amongst 30, 60, 90, and 120 s (V-HIIT), and when participants were allowed to autonomously self-select interval duration (A-HIIT) amongst 30, 60, 90, and 120 s. The self-selection process associated with the familiarization trial served to create an exercise session that provided autonomy support. Psychological responses of exertion, affect, and enjoyment were measured periodically throughout the sessions, while intention and enjoyment were measured after exercise.

Findings that HR and RPE increased over time in all experimental conditions is consistent with previous HIIT research designs utilizing workloads that were near maximum (Fleming et al., 2020; Kilpatrick et al., 2016; Martinez et al., 2015; Mcdaniel, 2019). The increase in HR from start to end of exercise sessions was most pronounced in the conventional HIIT session with a gradual increase of about 10 beats per minute, while HR was more variable over time in the varied and autonomous HIIT trials that included a mix of shorter and longer interval durations. Average HR values for the work segments across all trials were approximately 90% of peak HR, which provides evidence that the manipulation created sessions of exercise resulting in intense cardiovascular stress. Similar patterns were observed with RPE, with the largest increases observed within the conventional HIIT trial. Important to note is that all mean RPE responses stayed below the 'very hard' value of 17 on the Borg 6–20 scale and the average for all measurements was approximately 15 on the scale which is described as 'hard.' These numerical values for HR and RPE indicate that the trials were successful in creating an intense cardiovascular stimulus that can provide benefits for fitness, while producing exertion levels that were not extremely difficult.

Specific review of perceived exertion responses indicate notable differences across sessions. The autonomous HIIT session elicited lower RPE than the conventional HIIT session by about one half RPE unit. These findings indicate that even though the average physiological work was the same across sessions, the conventional session was perceived as more difficult than the autonomous session. This finding is similar to a previous study investigating exertion in imposed exercise compared to self-selected exercise (Hamlyn-Williams et al., 2014). This observation is perhaps predictable based on self-determination theory (SDT), in that participants realize more autonomy support when self-selecting sessions which could facilitate lower levels of exertion. While the differences in exertion match theory and existing data, the corresponding lower HR value for autonomous HIIT when compared to continuous HIIT is surprising because the overall physical stress should have been equivalent for all trials.

Findings related to affective valence indicate that pleasure associated with the three sessions of HIIT were not different from each other during work or recovery segments, but a trend towards significance was observed for the work segment through conventional HIIT producing less pleasure when evaluated by mean comparisons. Though affective valence was lower during exercise than baseline as evidenced by reductions of about one unit, each HIIT session remained positive during exercise and well above the neutral point on the scale at a level of pleasure that was 'fairly good' to 'good.' Evaluation of grand mean values for in-task affective valence revealed that autonomous HIIT exercise was more pleasureable than varied HIIT exercise by about one half FS unit. This finding is similar to other studies in moderate continuous exercise indicating that when participants self-select their intensity there is a more positive affective response (Haile et al., 2019). These affective valence responses are notable because of the possibility that in-task valence may predict future exercise behavior, (Chmielewski et al., 2016; Williams et al., 2016). While some research has argued against this possibility (Invanova, et al., 2020; Stork et al., 2018), there continues to be some sense that feeling good during exercise leads to being more physically active over time.

Consideration of in-task and post-exercise enjoyment responses reveal similar findings. Autonomous HIIT was more enjoyable than conventional and varied HIIT during exercise by about one half EES unit, while in the post-exercise period autonomous HIIT was more enjoyable than varied HIIT by about six PACES units. These findings suggest that the provision of autonomy in selecting HIIT exercise sessions results in more enjoyment. This pattern of results is consistent with what has been observed in cardiac rehabilitation environments which indicates that autonomy support leads to desirable motivations and more total exercise (Russell & Bray, 2010). The current findings are also consistent with research investigating school-based exercise programs, which indicate perceived increases in autonomy support facilitate increased intrinsic motivation, activity adherence and physical activity intention (Sfandyari et al., 2020). Within the context of this acute exercise study design, these findings suggest the absence of volition, perceived choice, and locus of control associated with prescribed and controlled exercise are less desirable from a motivational perspective.

A separate important consideration of the current design relates to post-exercise assessment of exercise intention. This finding indicates that intention for future exercise was higher for autonomous HIIT than varied HIIT and this outcome is predicted by self-determination theory based on previously described results for affective valence and enjoyment. Specifically, intention to engage in behaviors is linked to motivation and characteristics of enjoyment and interest (Papacharisis & Goudas, 2003; Zenko et al., 2016). Importantly, the participants in the current study were provided an opportunity to select the design characteristics in advance of the exercise session rather than during the session, which creates perhaps a different sense of autonomy support than would be experienced by providing real-time choices for interval duration during an exercise session.

One somewhat surprising finding across numerous measurement contexts was that the varied HIIT condition did not provide more favorable responses than conventional HIIT. The expectation of the researchers had been that autonomous HIIT would produce the most positive outcomes followed by varied HIIT, with conventional HIIT being the least positive. Instead it was found that varied HIIT often yielded responses that were equal to or less desirable than conventional HIIT. However, one potential explanation for the varied HIIT responses is that this session included two 120-s intervals, which has been demonstrated to produce less positive affect and enjoyment when compared to interval sessions comprised exclusively of 30-s and 60-s work intervals (Martinez et al., 2015). The findings from the current study suggest that the potential psychological benefits of variation that accompanied this HIIT session could not overcome the relatively less positive psychological responses that are produced by very long and taxing intervals that are perceived as more difficult and less pleasurable. Prolonged exposure to very intense intervals perhaps begins to create a physiological environment that more closely mimics intense continuous exercise, which is more metabolically stressful and likely to produce worsened affective responses (Ekkekakis et al., 2005). Importantly, psychological responses to the autonomous HIIT session was more positive than the varied HIIT session despite including 90-s and 120-s interval segments, and this could help explain the distinctly more positive responses to autonomous HIIT. When viewed more holistically, however, these findings together show that when individuals are given more autonomy support, the resulting exercise experience is more positive and desirable.

The current study presents with notable strengths including internal validity provided by way of tightly controlled exercise sessions that created a uniform testing environment that allowed for proper isolation of the dependent variables, and the inclusion of baseline affective valence measurements that allows for consideration of changes in psychological responses throughout the experimental trials. Additionally, the research question focusing on the impact of autonomy support on the psychological responses to exercise is novel and provides innovation. Limitations of the design are notable and point the way towards future studies that can build upon the current findings. One such limitation relates to the sample, which is relatively small and young. A larger and more diverse sample would have provided opportunities to consider moderator variables such as gender, age, fitness status, and physical activity experience that was not available in the current study. A second design limitation of note is that the manipulation to induce autonomy was only presumed to occur based on the manipulation but was not objectively assessed. A third consideration is that the utilization of a young and active sample perhaps more accustomed to vigorous exercise may have facilitated somewhat more positive responses than might be expected from less active or older individuals. Also noteworthy is that the design employed did not allow for a matching of shorter and longer interval durations across the varied and autonomous trials. Future studies might consider alternate ways to induce autonomy including providing participants the opportunity to self-select the individual segment duration of the intervals in real time, thus likely creating a more potent sense of autonomy but creating other challenges as it relates to the measurement scheme employed to match the exercise session. And finally, designs might consider creating autonomy by offering a choice of exercise modalities or by comparing indoor and outdoor exercise locations. One design feature in particular that would be desirable is the development of intervention conditions that vary on autonomy to determine impacts on chronic exercise behavior.

5. Conclusion

In summary, the current findings provide the basis for exercisers being given more autonomy when participating in HIIT exercise because such an approach yields more positive psychological responses. These findings make clear that autonomy is beneficial within HIIT exercise, but additional work can lead to a better understanding of how to best leverage autonomy support within the exercise experience. Future research may want to allow exercisers to adjust the duration and/or intensity of individual work and recovery segments, thus providing a different and perhaps more potent form of autonomy. Given that existing research makes clear that HIIT is generally well-tolerated, the provision of autonomy through any number of methodologies is likely to create an even better exercise experience that can positively impact health and related outcomes.

Declaration of competing interest

My colleagues and I declare that we have no conflicts of interest to report for the proposed research project.

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